

KAGGLE TITANIC · MODEL EVALUATION

Beyond accuracy: **trust your model**

Metrics, thresholds, validation, error analysis, and the evidence behind a model decision.

MEASURE

COMPARE

DECIDE

What you will be able to do

1

Read errors

Explain every cell in a confusion matrix.

2

Choose metrics

Match the metric to class balance and error cost.

3

Test stability

Use cross-validation instead of trusting one split.

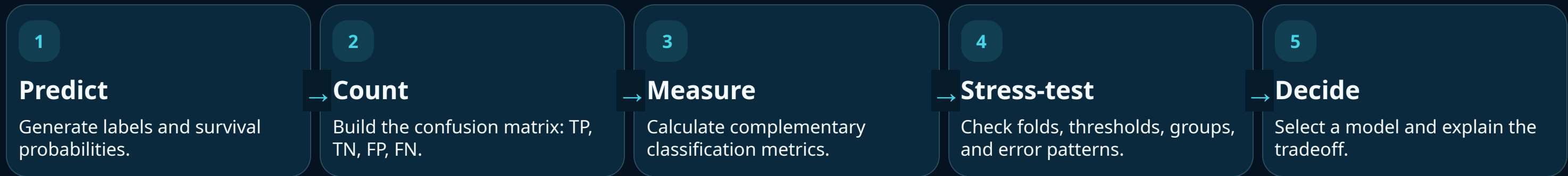
4

Make a decision

Compare models with evidence, not one leaderboard score.

The goal: know what your score means — and what it hides.

From predictions to a decision



Evaluation is a loop: errors suggest a new feature, cleaning decision, threshold, or model — then we validate again.

A label answers. A probability expresses confidence.

PREDICT()

0 or 1

Hard label

Used by Accuracy, Precision, Recall, Specificity, F1, and the confusion matrix.

PREDICT_PROBA()

0.00 → 1.00

Probability

Used by thresholds, ROC-AUC, Average Precision, Log Loss, Brier Score, and calibration.

Default rule: probability ≥ 0.50 becomes class 1. The threshold is a decision, not a law.

Every classification metric starts here

	Predicted 0	Predicted 1
Actual 0	TN Correct non-survivor	FP Predicted survival incorrectly
Actual 1	FN Missed a survivor	TP Correct survivor

Correct: TP + TN

Errors: FP + FN

False positive

The model says “survived,” but the passenger did not.

False negative

The model says “did not survive,” but the passenger did.

First question: are FP and FN equally costly for the real problem?

Accuracy is useful — with context

Accuracy

How often was the model correct overall?

$$(TP + TN) / ALL$$

Good when classes and error costs are reasonably balanced.

Balanced Accuracy

How well did the model perform across both classes?

$$(Recall + Specificity) / 2$$

More informative when one class is much more common.

Majority baseline: always predicting “did not survive” is already about 62% accurate on the Titanic training data.

Precision and Recall answer different questions

Precision

Of predicted survivors, how many truly survived?

$$TP / (TP + FP)$$

High precision: few false survival predictions.

Recall

Of all actual survivors, how many did we find?

$$TP / (TP + FN)$$

High recall: few survivors were missed.

Tradeoff: lowering the threshold usually raises Recall and lowers Precision; raising it usually does the opposite.

Measure the negative class and the balance

Specificity

Of all non-survivors, how many did we correctly reject?

$$TN / (TN + FP)$$

Also called the True Negative Rate.

F1 Score

What single score balances Precision and Recall?

$$2 \times (P \times R) / (P + R)$$

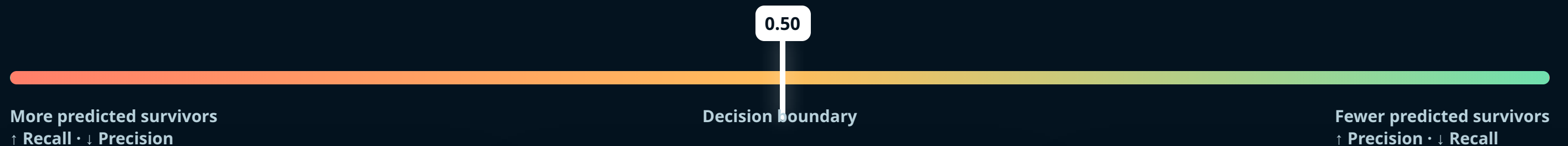
Useful when both FP and FN matter and the positive class is important.

Advanced robust summary: Matthews Correlation Coefficient (MCC) uses all four confusion-matrix cells and remains informative under imbalance.

Start with the decision you need to make

Question	Primary metric	Watch alongside it	Direction
Overall correctness?	Accuracy	Class balance	Higher is better
Uneven classes?	Balanced Accuracy	Per-class Recall	Higher is better
Avoid false alarms?	Precision	Recall	Higher is better
Avoid missed positives?	Recall	Precision	Higher is better
Balance Precision + Recall?	F1	Confusion matrix	Higher is better
Rank positives above negatives?	ROC-AUC / AP	Curve shape	Higher is better
Trust probability quality?	Log Loss / Brier	Calibration plot	Lower is better

The same model can make different decisions



Threshold 0.30

More positive predictions. Useful when missing a positive is costly.

Threshold 0.50

Common default. It is not automatically the optimal operating point.

Threshold 0.70

Fewer positive predictions. Useful when false positives are costly.

Choose the threshold on validation data using a stated objective. Never tune it against hidden Kaggle answers.

ROC-AUC checks every threshold



ROC curve: Recall versus False Positive Rate across thresholds.

0.50

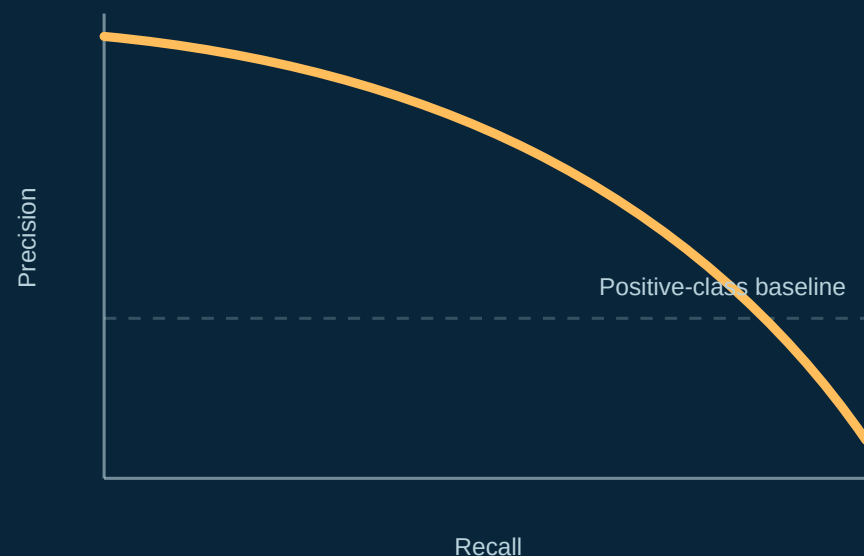
Random ranking

1.00

Perfect ranking

Caution: ROC-AUC can look optimistic when the positive class is extremely rare.

Precision–Recall focuses on the positives



PR curve: shows how Precision changes as Recall increases.

Average Precision (AP)

Summarizes the curve; higher is better.

Best use

When positives are rare or positive-class performance is central.

Baseline

The positive-class prevalence, not always 0.50.

Are the probabilities honest?

LOG LOSS

Penalizes confident mistakes

A wrong prediction at 0.99 confidence hurts far more than one at 0.55.

↓ Lower is better

BRIER SCORE

Squared probability error

Measures the average squared gap between probability and the 0/1 outcome.

↓ Lower is better

CALIBRATION

Confidence matches reality

Among passengers predicted near 0.70, roughly 70% should be positive.

Closer to diagonal is better

Important: a model can rank passengers well (high AUC) while producing poorly calibrated probabilities.

One split may be lucky. Use several.

VALID	TRAIN	TRAIN	TRAIN	TRAIN
TRAIN	VALID	TRAIN	TRAIN	TRAIN
TRAIN	TRAIN	VALID	TRAIN	TRAIN
TRAIN	TRAIN	TRAIN	VALID	TRAIN
TRAIN	TRAIN	TRAIN	TRAIN	VALID

Every row is used for validation once and for training four times.

Mean score

Expected performance across folds.

Standard deviation

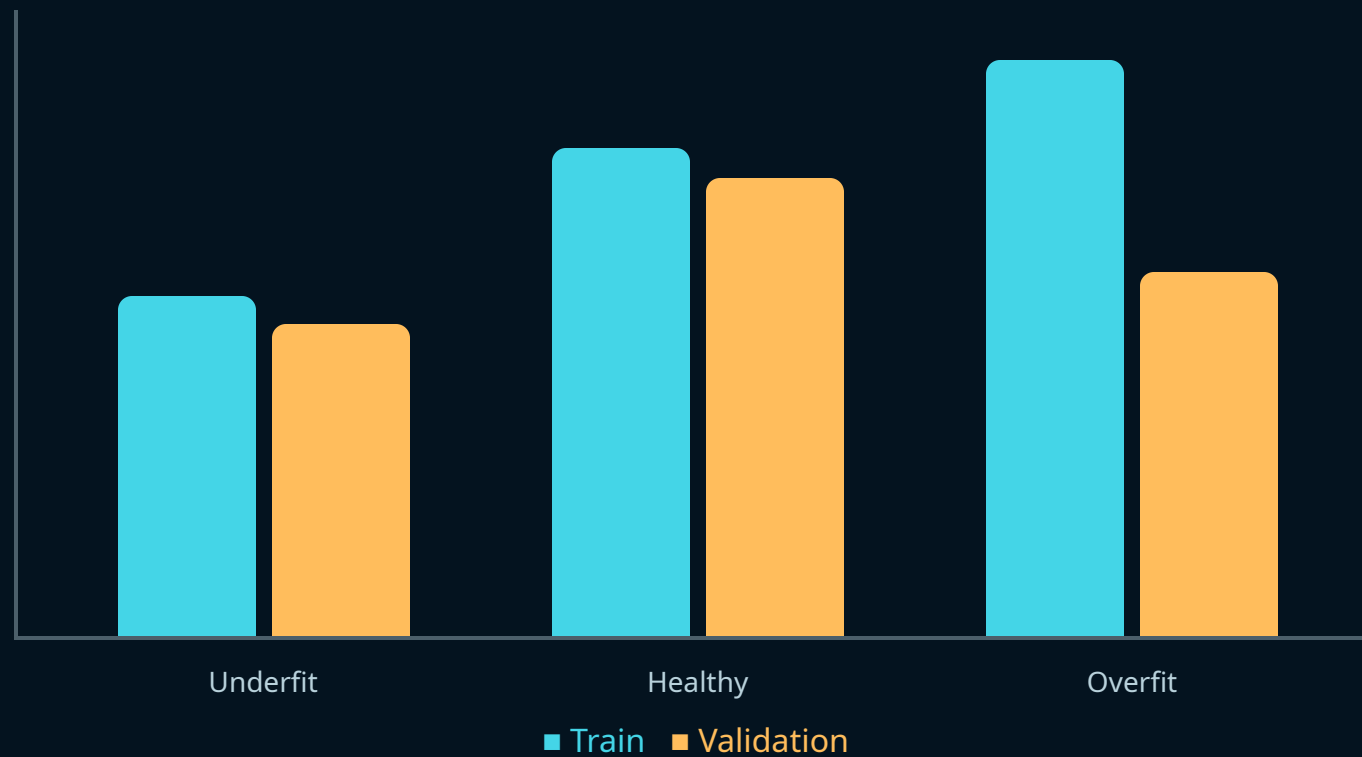
How sensitive the result is to the split.

Stratification

Preserves class ratios in each fold.

Pipeline required: preprocessing must be refit inside every fold.

Compare training and validation



Underfitting

Both scores are low: model or features are too simple.

Healthy fit

Both scores are useful and the gap is small.

Overfitting

Training is high, validation drops: the model memorized details.

Learning curve

Shows whether more data may improve generalization.

An average can hide weak groups

Calculate metrics for meaningful groups such as **Sex** and **Pclass**.

Coverage

Is every group large enough to interpret?

Error rates

Do FP or FN rates differ sharply?

Data quality

Is missingness concentrated in one group?

Small groups are noisy: always report their sample size with the metric.



Public feedback is not the final answer

Public leaderboard		VISIBLE NOW
1	Model C	0.806
2	Model A	0.799
3	Model B	0.792
Uses only part of the hidden answers and gives iterative feedback.		

Private leaderboard		FINAL
1	Model A	0.803
2	Model B	0.795
3	Model C	0.781
Uses the remaining hidden answers and determines the final ranking.		

Leaderboard overfitting: repeated choices based on the public score can make you specialize to that subset. Trust cross-validation and limit submissions.

Build a model scorecard

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.80	0.78	0.73	0.75	0.86
Decision Tree	0.78	0.76	0.68	0.72	0.80
Random Forest	0.82	0.82	0.71	0.76	0.85

Performance

Which metrics match the objective?

Stability

Are cross-validation scores consistent?

Simplicity

Can we explain and maintain the model?

Illustrative values: replace them with your notebook results. The “winner” depends on the goal and acceptable error tradeoff.

Evaluate like a scientist

- 1 Evaluate hard labels**
Confusion matrix, Accuracy, Balanced Accuracy, Precision, Recall, Specificity, F1, and MCC.
- 2 Evaluate probabilities**
ROC-AUC, Average Precision, Log Loss, Brier Score, and calibration.
- 3 Test three thresholds**
Compare 0.30, 0.50, and 0.70.
- 4 Run 5-fold cross-validation**
Report mean and standard deviation.
- 5 Compare models and groups**
Choose one model and defend your decision.

Deliverables

- One metric scorecard
- Confusion matrix
- ROC and PR curves
- Threshold comparison
- Cross-validation summary
- Three-sentence recommendation

PART III COMPLETE

Measure what matters.

A trustworthy model is accurate enough, stable, understandable, and evaluated for the decision it supports.

QUESTIONS?